

AARADHYA DEV TAMRAKAR

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PROFILE

Electronics, Communication and Information Engineering student at IOE, Kathmandu Engineering College (Year IV/I). Focused on Artificial Intelligence, robotics, and embedded systems. Experienced building ML pipelines (classification, regression, tree-based ensembles, agentic Text-to-SQL systems), deploying models via FastAPI and Docker, and developing Android applications with Kotlin/Jetpack Compose. Currently a Fellow at the Fusemachines AI Fellowship (2026, Weeks 1–6 complete) and NSSR DataCamp Fellow (Cohort 2, 2026). Interested in applying AI to real-world sensor systems, robotics control, and edge intelligence.

EDUCATION

Bachelor of Engineering (B.E.) — Electronics, Communication and Information Engineering *Expected: January 2027*
Institute of Engineering (IOE), Kathmandu Engineering College | Year IV / Part I

Coursework: Machine Learning · Signals and Systems · Digital Electronics · Microprocessors · Control Systems · Communication Systems · DBMS · Project Management · Propagation & Antenna

EXPERIENCE

Vice Chair

2026 – Present

IEEE KEC KTM Student Branch

- **Strategic planning:** Spearheading high-level decision-making to scale technical initiatives and member engagement.
- **Institutional relations:** Directing cross-functional executive committees and managing the broader engineering community ecosystem.

Event Manager

2026 – Present

Electronics Project Club (EPC), Kathmandu Engineering College

- Managing logistics and coordination for engineering events and technical workshops.

Vice Secretary

2025 – 2026

IEEE KEC KTM Student Branch, Kathmandu Engineering College

- **Operations:** Managed official documentation, meeting minutes, and internal communication frameworks, streamlining branch operations.
- **Event delivery:** Orchestrated logistics for technical workshops and IEEE programs, significantly increasing student participation.
- **Volunteer management:** Supported the executive committee in coordinating volunteers for seamless event execution.

Resource Manager

2024 – 2026

Electronics Project Club (EPC), Kathmandu Engineering College

- **Logistics:** Managed scheduling and resource allocation for engineering workshops and technical events across multiple departments.

FELLOWSHIPS & RECOGNITIONS

Fuse AI Fellow — Fusemachines AI Fellowship 2026

Active

Fusemachines

- **Admission:** Passed entrance examination (March 2026) covering linear algebra, calculus, probability, Python, and ML.
- **REST API:** Built a containerized customer REST API: FastAPI + PostgreSQL + Docker (4-layer architecture, asyncio concurrency).
- **Text-to-SQL agent:** Developed a five-stage agentic pipeline (Planner → Generator → Validator → Executor → Summarizer) achieving 100% execution success and 100% accuracy on a 50-question benchmark; basis for IT4D 2026 concept paper.

- **Churn & CLV modeling:** End-to-end pipeline with Logistic Regression / Ridge / SGD classifiers; Stratified 5-fold CV ROC-AUC 0.841 ± 0.005 ; threshold-tuned for top-200 high-risk segment; Papermill automation.
- **Tree-based ensemble:** Random Forest + XGBoost with ImbPipeline/SMOTE (zero leakage), SHAP explainability, GridSearch tuning, and Joblib serialization.
- **Probabilistic models:** Bayesian estimation (MLE/MAP/full Bayes), Dirichlet-multinomial inference, Gaussian process regression, probabilistic graphical models, and Bayesian logistic regression via PyMC/ArviZ/pgmpy.

NSSR DataCamp Fellow — Cohort 2

2026 – Present

Nepalese Society of Student Researchers

- **Selection:** Competitively selected (April 2026) for a sponsored DataCamp Premium license; one of a limited cohort of Nepalese student researchers.
- **GenAI applications:** Engineered 6 applied GenAI solutions across product strategy and data workflows — deploying advanced prompt engineering and LLM constraint handling.
- **Data automation:** Reduced manual data preparation overhead by applying generative AI for automated null filtering, deduplication, and schema normalization.
- **Certifications:** SQL Associate + Python Data Associate (targeting June 2026); covering Data Manipulation in PostgreSQL, Pandas, Statistical Thinking, and ML with scikit-learn.

PROJECTS

Text-to-SQL Agentic Pipeline *Python · FastAPI · Streamlit · GPT-4o-mini · PostgreSQL*

- **System design:** Production-grade automated Text-to-SQL over classicmodels PostgreSQL for a 50-question benchmark.
- **Task 3:** Streamlit chat UI + vanilla Python prompt chaining with SQL safety validation (blocks DML/DDL) and query execution logging to JSON.
- **Task 4:** FastAPI modular planner-generator-validator-executor-summarizer state machine, up to 3 retries. 100.0% execution success and 100.0% accuracy across all 50 questions with zero retries.

Telco Churn Tree-Based Ensemble Pipeline *Python · XGBoost · Random Forest · SHAP · ImbPipeline · Joblib*

- **Pipeline:** End-to-end classification and regression on Telco Customer Churn (7,043 rows, ~27% positive rate); exposed Accuracy Trap with naïve baseline.
- **Leakage prevention:** ImbPipeline restricts SMOTE to training folds only; ColumnTransformer for mixed dtypes.
- **Explainability:** SHAP global summary + local waterfall/force plots; retention recommendation for specific at-risk customer.

Gesture-Controlled Self-Balancing Robot (GCSBR) — Minor Project *Computer Vision · Arduino · MPU6050 · Stepper Motors · Android · MATLAB*

- **Evaluation:** Described as 'major project level' by examiner.
- **Firmware V6.9.1:** Watchdog timer, atomic writes, differential ramp, integrator bleed, PID stabilization. MATLAB simulation → hardware deployment pipeline.
- **Interface:** Android gesture-control app (MediaPipe, CameraX, HC-05 BT); dual-hand gesture control; DRV8825 drivers on CNC Shield for real-time actuation.

Edge AI Stability Detection System *Python · Scikit-learn · FastAPI · Joblib*

- ML system predicting platform stability from simulated IMU sensor data (tilt_x, tilt_y, angular velocity). 10,000-sample synthetic dataset; Random Forest — 99.8% test accuracy. REST API via FastAPI for real-time predictions.

Alpha Android Super-App *Kotlin · Jetpack Compose · Material3 · DataStore · Apache POI*

- Modular Android super-app (SDK 36): SBR gesture control, multi-mode calculator (STD/PROG/LOGIC), budget tracker with eSewa Gmail parser, custom Astronomus font, full Material3 design system.

SysOptimizer — Windows Optimization Tool *Python · CustomTkinter · PyInstaller · WMI · PowerShell*

- Standalone .exe with animated ring gauges, sparklines, process table. v4/v5: Power Plan switcher, RAM flush, Background Bloat panel (14 services/tasks), Startup Scanner.

Custom Processor FSM Design *VHDL · Vivado 2023.2 · FSM · Datapath · FPGA*

- VHDL implementation of custom processor datapath and FSM supporting GCD and exponentiation operations. Simulated and verified in Vivado 2023.2.

Probabilistic Modeling Suite (Fusemachines Week 6) *Python · PyMC · ArviZ · pgmpy · scikit-learn · Pandas*

- End-to-end Bayesian analysis suite: MLE/MAP/full Bayes estimation, sequential belief updating, Dirichlet-multinomial inference, multivariate Gaussian conditioning, probabilistic graphical models, Gaussian process regression, and PyMC Bayesian logistic regression with ArviZ posterior diagnostics. Artifact: telco_bayes_lr_v1.pkl.

Nexus — Personal AI Operating System *React (Vite) · FastAPI · PostgreSQL · ChromaDB · httpx · asyncio*

- Project-centric AI operating system unifying Claude, Gemini, ChatGPT, and Ollama into a single workspace via parallel fan-out routing (asyncio.gather). FastAPI backend + React frontend + PostgreSQL history + ChromaDB RAG. Day 1 skeleton operational: multi-model parallel responses at localhost:5173.

SKILLS

Programming: Python, C, C++, Kotlin, SQL (PostgreSQL / SQLite), VHDL

Machine Learning: scikit-learn, NumPy, Pandas, XGBoost, Random Forest, Logistic/Ridge/SGD Classifiers, Lasso, ElasticNet, SMOTE (ImbPipeline), SHAP, GridSearchCV, Stratified K-Fold CV, Joblib, PyMC, ArviZ, pgmpy, Bayesian Inference (MLE/MAP/full Bayes), Gaussian Process Regression

Android Development: Kotlin, Jetpack Compose, Material3, DataStore, CameraX, MediaPipe, Apache POI

Deployment: FastAPI, REST APIs, Docker, Docker Compose, PostgreSQL (containerized), Streamlit, asyncio, Papermill, Joblib

Embedded Systems: Arduino, MPU6050, CNC Shield, DRV8825, Stepper Motors, HC-05 BT, UART, FPGA, Vivado 2023.2

AI / Local LLM: Ollama, AnythingLLM, ipex-llm, DeepSeek R1 7B (Intel Arc iGPU), Prompt Chaining, Agentic Query Systems

Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, MATLAB, LaTeX / Overleaf, SSMS

Other: Event Coordination, Project Management, Public Relations, Microsoft Office, Canva, Blender (Beginner)

TECHNICAL ACTIVITIES & CERTIFICATIONS

- IEEE WIE Nepal Section LaTeX Workshop 2026 — KEC Makerspace (May 5–6, 2026). Certificate of Participation. Skills: LaTeX, Technical Writing, Beamer, Document Formatting.
- NepaTronix 3-Day Drone Training — May 2–4, 2026. Backed by Nepal Police / Drone Association of Nepal / IITM Pravartak. Skills: Drone Technology, UAV Operations, Flight Safety Protocols.
- IEEE Conference Leadership Workshop 2026 — IEEE Nepal Section. Skills: Event Management, Leadership, Conference Organization.
- IEEEExtreme 19.0 Participant (Team: ShadowXTREME) — Oct 25, 2025. Skills: Competitive Programming, Algorithms, Data Structures, C++.
- AWS Cloud Computing Workshop — KEC IT Club (2025). Hands-on with EC2, S3, and cloud deployment fundamentals.
- PreXtreme Competitive Programming Workshop — IEEE Pulchowk (2025).
- Git & GitHub Workshop — KEC IT Club (2024).

Active member: IEEE KEC KTM Student Branch · Electronics Project Club · KEC Maker's Space · KEC Music Club

Last Updated: June 10, 2026